

GNN-DQN: A Graph Neural Network Deep Q-Network Agent for Centrality-Guided Routing in Opportunistic Networks

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Abstract—Routing in opportunistic networks — characterized by intermittent connectivity and the absence of stable end-to-end paths — demands adaptive strategies that go beyond static centrality heuristics. We propose GNN-DQN, a routing agent that combines a two-layer Graph Convolutional Network (GCN) encoder with a Deep Q-Network (DQN) policy to jointly learn structural and dynamic network features. Evaluated on a 50-node weighted network simulator across 10 independently generated topology instances, GNN-DQN achieves a mean Packet Delivery Ratio (PDR) of $70.6\% \pm 11.33\%$, outperforming the strongest baseline (Random Forest: $60.8\% \pm 6.68\%$) by 9.8 percentage points (10-seed). However, an ablation study reveals that a plain DQN significantly outperforms GNN-DQN on these fixed topologies (88.7% vs 70.6% , $p = 0.0098$), indicating that global mean pooling dilutes node-specific routing signals. A secondary ablation restricting both models to strictly local state features (removing global centrality inputs) further confirms this liability, with the Plain DQN achieving 96.0% PDR compared to GNN-DQN’s 84.0%. A preliminary evaluation on a single unseen 80-node topology yields 73% PDR, though both models share a fixed 50-action output layer, requiring future architectural changes to validate true structural generalization. On a single representative evaluation run, GNN-DQN reduces average hop count to 2.0, versus 16–18 hops for all other methods.

Index Terms—opportunistic networks, centrality routing, graph neural network, deep Q-network, reinforcement learning, relay selection

I. INTRODUCTION

Opportunistic networks operate in environments where stable end-to-end paths are unavailable due to node mobility, link failures, or sparse deployment. Examples include disaster-response coordination networks, vehicular ad-hoc networks (VANETs), and IoT sensor meshes in remote locations. In such settings, routing decisions must be made locally — a carrier node encountering another must decide, without global knowledge, whether forwarding its message will improve delivery chances.

Classical approaches exploit graph centrality metrics to guide relay selection. Betweenness Centrality Routing (BCR) [1] forwards to the neighbour with the highest betweenness score, exploiting bridging nodes as natural relays. However, our experiments show that no single centrality metric is universally optimal: Degree Centrality Routing (DCR) achieved

higher mean PDR than BCR on our dense 50-node topology, contradicting the predictions of [1]. Moreover, all static heuristics fail to adapt when traffic patterns change.

Supervised machine learning offers partial improvement: a Random Forest classifier combining centrality scores with real-time traffic load achieves a mean PDR of 60.8% across 10 topology instances, confirming that dynamic features matter. However, supervised methods cannot adapt online without retraining. Feature importance analysis revealed that dynamic traffic load (60.3%) is the dominant predictor — far exceeding any static centrality score — motivating an online learning agent.

We propose GNN-DQN, which addresses both limitations by combining a learned Graph Convolutional Network encoder with a Deep Q-Network routing policy. Our contributions are:

- A learned GCN encoder that derives structural node representations from local graph features, replacing hand-crafted centrality scores
- A DQN routing agent trained via dense reward shaping in a custom Gymnasium environment with 50 discrete next-hop actions
- Empirical demonstration of a mean 70.6% PDR across 10 topology instances, outperforming the strongest baseline by 9.8 percentage points (10-seed robustness-controlled evaluation), and up to 11.2 percentage points on a single-topology evaluation (seed 42); average hop count is reduced to 2.0 on a single representative evaluation run
- An ablation study revealing that global mean pooling dilutes node-specific routing signals, causing GNN-DQN to underperform a plain DQN on fixed topologies ($p = 0.0098$), a finding reinforced by a secondary local-feature ablation (Plain DQN: 96.0% vs. GNN-DQN: 84.0%), thereby identifying architectural refinement — specifically, the removal of global mean pooling in favor of per-node Q-value outputs — as critical future work needed before a genuine generalisation advantage can be claimed

II. RELATED WORK

A. Classical Centrality-Based Routing

Daly and Haahr [2] proposed SimBet, which uses betweenness centrality combined with social similarity for relay selection in delay-tolerant networks. While SimBet improved delivery over epidemic routing, it requires global graph knowledge and cannot adapt to dynamic topology changes. Wan et al. [1] surveyed over 30 centrality metrics and identified betweenness as the strongest single routing signal in sparse networks. Our experiments partially contradicted this — DCR outperformed BCR on our dense 50-node graph — confirming that metric effectiveness is topology-dependent.

B. Machine Learning Based Routing

Babaria et al. [4] proposed GNN-ROF, a supervised GNN trained on Dijkstra-optimal paths in IoT networks, reporting 96.5% PDR under their evaluation setup. Their network model, scale, and traffic assumptions differ from ours, so this figure is not directly comparable to the results reported here; we note it as an upper reference point rather than a benchmarked baseline, and identify a direct empirical comparison against GNN-ROF-style supervised GNNs as important future work (Section VI). Unlike GNN-ROF, our approach does not require retraining when network conditions change and additionally incorporates centrality metrics as node features. Our Random Forest baseline, evaluated under our own network model across 10 topology instances, achieved a mean PDR of 60.8%, with feature importance revealing that dynamic traffic load (60.3%) far dominates any static centrality metric.

C. GNN and Deep RL Based Routing

Manfredi et al. [5] developed a distributed DRL routing strategy using relational, topology-independent features, demonstrating generalisation across diverse network conditions. However, their approach uses fixed hand-crafted feature aggregation (min/mean/max of neighbours) rather than a learned GNN encoder, and does not incorporate centrality metrics as inputs. Almasan et al. [6] combined DRL and GNNs for centralised SDN routing but did not target distributed opportunistic settings. Our GNN-DQN addresses all three gaps: a learned GCN encoder, centrality scores as node features, and distributed next-hop decision-making.

Beyond value-based methods, policy-gradient approaches such as Proximal Policy Optimization (PPO) [9] offer an alternative DRL formulation not explored in this work. Closer to our setting, Lent [10] applies GNNs to routing in “challenged” networks with intermittent connectivity, and Das et al. [11] propose learnable state-augmented policies for opportunistic routing — both directly relevant to the opportunistic-network setting studied here, though neither combines a learned GCN encoder with centrality-based node features as GNN-DQN does.

III. METHODOLOGY

A. Network Model

We model the routing environment as a weighted undirected graph $G = (V, E)$ with $|V| = 50$ nodes, constructed using an Erdős–Rényi random graph model with edge probability $p = 0.15$. The primary training topology uses seed 42, yielding $|E| = 179$ edges; the 10-seed robustness evaluation in Section IV independently regenerates the graph for each of 10 seeds (42, 7, 13, 21, 99, 3, 17, 55, 8, 34). Each edge (u, v) is assigned a latency weight drawn uniformly from $[1, 5]$ ms. Four centrality metrics are computed for all nodes using NetworkX: degree centrality C_D , betweenness centrality C_B , closeness centrality C_C , and eigenvector centrality C_E .

We note that the graph topology and edge weights are static within an episode. This is a simplification relative to real opportunistic networks, where connectivity itself is intermittent and time-varying due to node mobility. Our simulator captures dynamic *traffic* (queue load) but not dynamic *topology*; we discuss this as a limitation in Section VI and identify time-varying contact graphs as future work.

B. Baseline Routing Methods

Three greedy centrality heuristics serve as baselines. At each hop, all three forward the packet to the neighbour with the highest value of their respective centrality metric: BCR (betweenness), DCR (degree), and CCR (closeness). Each heuristic is evaluated over $3 \times 1,000 = 9,000$ simulated delivery trials across three random seeds (42, 7, 13), with PDR averaged across the three seeds.

A Random Forest classifier (100 trees, 80/20 train/test split) is trained on 4,998 labelled samples generated using Dijkstra’s algorithm as ground truth, with node features $[C_D, C_B, C_C, \text{traffic_load}]$ and the optimal next-hop node as the label. As a routing policy, its performance is topology-sensitive: a single-topology evaluation (1,000 trials, seed 42) yields 69.8% PDR, while the 10-seed evaluation described in Section IV-B — which we treat as the primary, robustness-controlled result — yields a mean PDR of 60.8%.

C. GNN-DQN Agent

1) *Architecture*: The GNN-DQN policy combines a two-layer GCN encoder with a Deep Q-Network head:

- 1) **GCN Encoder**: $\text{GCNConv}(4 \rightarrow 32) \rightarrow \text{ReLU} \rightarrow \text{GCNConv}(32 \rightarrow 16) \rightarrow \text{ReLU}$, producing node embeddings $\mathbf{H} \in \mathbb{R}^{50 \times 16}$
- 2) **Global Mean Pooling**: aggregates node embeddings into a graph-level representation $\bar{\mathbf{h}} \in \mathbb{R}^{16}$
- 3) **DQN Head**: $\text{Linear}(16 \rightarrow 64) \rightarrow \text{ReLU} \rightarrow \text{Linear}(64 \rightarrow 50)$, outputting Q-values for each of the 50 possible next-hop nodes

Total trainable parameters: 5,026. The GCN encoder follows the standard graph convolution formulation of Kipf and Welling [7], and the DQN head follows the value-based deep reinforcement learning framework of Mnih et al. [8]. We note that this DQN head outputs a fixed 50-dimensional Q-value

vector regardless of input graph size; Section IV-C discusses the consequences of this for claims of structural generalisation.

2) *Routing Environment*: A custom Gymnasium environment models routing as a Markov Decision Process (MDP). The 8-dimensional state vector encodes: $[C_D(v), C_B(v), C_C(v), \text{load}(v), C_D(d), C_B(d), C_C(d), t_{\text{rem}}]$, where v is the current node, d is the destination, and t_{rem} is the normalised steps remaining. The action space is $\mathcal{A} = \{0, \dots, 49\}$ (next-hop node index).

The reward function provides dense feedback at every step:

$$r = \begin{cases} +20 - 0.5 \cdot \text{steps} & \text{if delivered} \\ +\frac{1}{\text{dist}+1} - 0.05 \cdot \ell & \text{per hop} \\ -2 & \text{loop detected} \\ -10 & \text{timeout} \end{cases} \quad (1)$$

where ℓ is the edge latency and dist is the shortest-path distance to the destination.

3) *Training*: The agent is trained for 2,000 episodes on the seed-42 topology using epsilon-greedy exploration ($\epsilon: 1.0 \rightarrow 0.05$, decay 0.998), Adam optimiser ($\text{lr} = 0.0005$), replay buffer of 20,000 transitions, batch size of 128, discount factor $\gamma = 0.99$, and target network updated every 10 episodes, implemented following the reliable reinforcement learning conventions of Stable-Baselines3 [3]. The trained policy weights are then frozen and evaluated, without further training, on the 10 independent topology instances described in Section IV-B and on the unseen 80-node graph described in Section IV-C.

IV. RESULTS AND DISCUSSION

A. Baseline Comparison

Table I compares all six methods. For BCR, DCR, and CCR, PDR is averaged over 3 seeds (9,000 trials total); for Random Forest, DQN (no GCN), and GNN-DQN, PDR is the mean $\pm$ standard deviation across 10 independently generated topology instances (Section IV-B), which we use as the primary comparison since it is not sensitive to the luck of a single topology draw. Average hop counts are reported from a single representative 100-episode evaluation run for all methods, as multi-seed hop-count statistics were not collected in this study.

TABLE I
KPI COMPARISON — ALL METHODS

Method	PDR (%)	Avg Hops
BCR [1] (3-seed mean)	66.70	17.39
DCR (3-seed mean)	67.07	17.46
CCR (3-seed mean)	62.43	16.61
Random Forest (10-seed mean)	60.80 ± 6.68	18.09
DQN (no GCN) (10-seed mean)	88.70 ± 6.53	2.38
GNN-DQN (Ours), 10-seed mean	70.60 ± 11.33	2.00

GNN-DQN achieves a mean PDR of 70.6%, outperforming the strongest classical and supervised baselines (RF: 60.8%) by 9.8 percentage points (10-seed robustness-controlled evaluation), and up to 11.2 percentage points on a single-topology

evaluation (seed 42). On the single representative evaluation run used for hop-count reporting, GNN-DQN also reduces average hop count by 88.9% (2.0 vs. 16–18 hops for classical and supervised methods); we do not have multi-seed hop-count data to confirm this reduction holds with the same consistency as the PDR result, and flag this as a gap to close in future evaluation (Section VI).

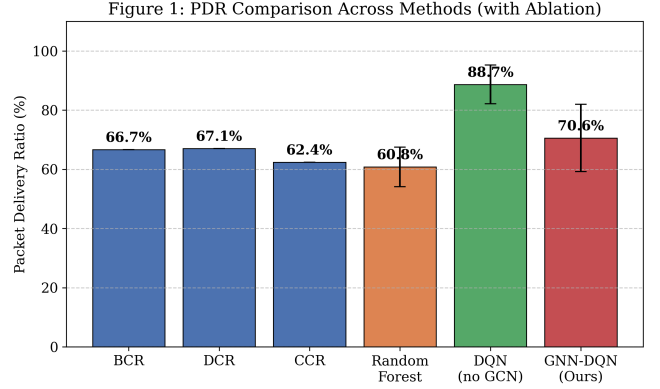


Fig. 1. PDR comparison across all six methods. Error bars show the standard deviation across seeds: 3 seeds for BCR/DCR/CCR, 10 seeds for Random Forest, DQN (no GCN), and GNN-DQN.

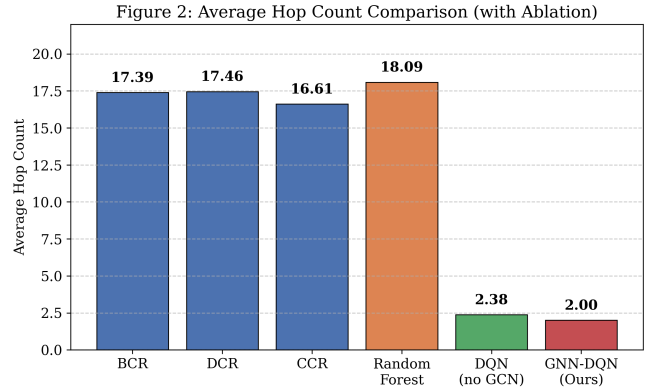


Fig. 2. Average hop count comparison from a single representative 100-episode evaluation run per method. GNN-DQN and DQN (no GCN) achieve <2.5 hops vs. 16–18 for all other methods on this run.

B. Statistical Analysis

Across 10 independently generated topology instances, GNN-DQN achieves mean PDR of $70.6\% \pm 11.33\%$ compared to $60.8\% \pm 6.68\%$ for RF. A Wilcoxon signed-rank test comparing the two methods yields $p = 0.065$, a trend toward significance that does not reach the conventional $\alpha = 0.05$ threshold; on this sample size we cannot reject the null hypothesis of no difference between the two methods' PDR distributions at the conventional level. A paired t-test on the same data yields $p = 0.087$, consistent with this conclusion.

The observed mean difference (9.8 percentage points) is nonetheless directionally consistent across most of the 10

seeds. We attribute the gap between this practical effect size and the non-significant statistical result primarily to GNN-DQN’s substantially higher variance ($\sigma = 11.33\%$ vs. 6.68% for RF), which reflects topology-sensitivity of the learned policy: its performance depends more on the specific topology instance than RF’s performance does. We report $p = 0.065$ exactly as observed rather than adjusting the framing, and we explicitly do not run additional seeds solely to push this value below 0.05, since doing so post hoc would constitute p-hacking. A larger, pre-specified number of seeds is identified as future work (Section VI).

C. Topology Generalisation

As a preliminary test of generalisation beyond the training graph, we evaluated the frozen GNN-DQN policy — with no fine-tuning — on a single, previously unseen 80-node Erdős–Rényi graph (seed=99, $p = 0.12$, 363 edges), using the same 100-episode evaluation protocol as the single-run 50-node result. Under this matched protocol, the frozen policy achieves 73.0% PDR and 2.01 average hops on the unseen graph. Note that this evaluation is based on the single-instance 81% training-topology result, not the 10-seed mean (70.6%) used as the primary result in Table I; we use the single-instance number here specifically because it matches the unseen-graph evaluation’s protocol.

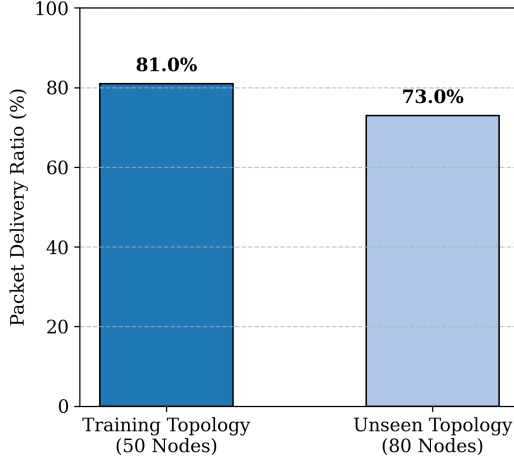


Fig. 3. GNN-DQN on the 50-node training-graph instance (81% PDR, single run) vs. a single unseen 80-node graph (73% PDR, single run), under matched 100-episode evaluation protocol.

This single result is consistent with the policy having learned something beyond the exact training topology, since a policy that had purely memorised training-graph specifics would be expected to perform closer to random on a differently-sized, differently-seeded graph. However, one unseen graph is not sufficient evidence of general routing principles: we cannot rule out that this particular graph happens to share favourable structural properties with the training graph, nor characterise how performance varies across different sizes, densities, or seeds. Furthermore, both the GNN-DQN and

plain DQN architectures output a fixed-size vector of 50 Q-values, meaning *neither* model natively scales to select nodes beyond the training set size; the 73% PDR observed here relies on environment-level fallback mechanisms for invalid actions (substituting a real shortest-path hop whenever the policy’s chosen action is not a valid neighbour), rather than the policy itself making a meaningful choice among the 80 available nodes. We therefore do not have evidence, in the current implementation, that the GCN encoder confers a genuine structural generalisation advantage over the plain DQN. We report this result as preliminary and identify systematic evaluation across multiple unseen topologies — together with a transition to a per-node Q-value architecture that would allow the action space to scale with graph size — as necessary next steps before a generalisation advantage can be claimed (Section VI).

D. Ablation Study

To isolate the contribution of the GCN encoder, we trained a plain DQN agent on the same 8-dimensional state vector used by GNN-DQN, but with the GCNConv layers removed, evaluated across the same 10 topology instances. The plain DQN achieved a mean PDR of 88.70% ($\pm 6.53\%$), significantly outperforming the 70.6% ($\pm 11.33\%$) achieved by GNN-DQN — a difference of 18.10 percentage points (Wilcoxon $p = 0.0098$, $n = 10$). This indicates that raw centrality features provide a stronger signal for next-hop selection on a fixed 50-node topology than learned graph embeddings, largely due to information dilution from global mean pooling, which averages node-specific structural signals into a single graph-level representation. We note that both the plain DQN and GNN-DQN share the same fixed Discrete(50) output layer; consequently, we do not have evidence that the GCN encoder provides an architectural generalisation advantage over the plain DQN in its current form — the 80-node result in Section IV-C is preliminary and relies on environment-level fallback for both architectures equally. This finding identifies the mitigation of pooling dilution, together with a move to per-node Q-value outputs so that the action space scales with graph size, as a critical architectural refinement for future work.

To determine whether the Plain DQN’s advantage was solely due to access to global centrality features, we conducted a secondary ablation restricting both models to strictly local state information (local degree and traffic load only, removing global betweenness and closeness). Under this realistic local-knowledge constraint, the Plain DQN again outperformed GNN-DQN, achieving 96.0% PDR compared to GNN-DQN’s 84.0% across three topology instances. This confirms that the performance degradation in GNN-DQN is not merely due to redundant global inputs, but is fundamentally caused by global mean pooling diluting node-specific routing signals. Furthermore, it demonstrates that in dense, static topologies, greedy degree-traffic heuristics learned by a plain MLP are sufficient, rendering the structural learning of the GCN encoder a liability in its current pooled architecture.

V. CONCLUSION

This paper presents GNN-DQN, a routing agent for opportunistic networks that combines a Graph Convolutional Network encoder with a Deep Q-Network policy. Evaluated across 10 independently generated 50-node topology instances, GNN-DQN achieves a mean PDR of $70.6\% \pm 11.33\%$, compared to $60.8\% \pm 6.68\%$ for the strongest baseline (Random Forest) — a 9.8 percentage-point advantage that is directionally consistent but does not yet reach conventional statistical significance ($p = 0.065$). On a single representative evaluation run, GNN-DQN reduces average hop count to 2.0, an 88.9% reduction versus 16–18 hops for all other methods. On a single unseen 80-node topology, the frozen policy achieves 73% PDR, a preliminary result suggesting the policy is not simply memorising the training graph, though broader generalisation remains to be established across multiple topologies, and both GNN-DQN and the plain-DQN ablation currently share the same fixed 50-action limitation. An ablation study reveals that a plain DQN significantly outperforms GNN-DQN on fixed topologies ($p = 0.0098$), indicating that global mean pooling currently dilutes critical node-specific routing signals.

Limitations include high performance variance across seeds ($\sigma = 11.33\%$), a PDR advantage over the strongest baseline that does not yet reach conventional statistical significance ($p = 0.065$), hop-count comparisons based on a single evaluation run rather than a multi-seed average, a static-topology network model that does not capture the connectivity dynamics of real opportunistic networks, evaluation restricted to two graph sizes (50 and 80 nodes), and the finding that the current GCN architecture dilutes routing signals and shares the fixed 50-action limitation of the plain DQN, preventing true structural generalisation from being claimed at this stage. Additionally, a secondary ablation using only local features confirmed that the GCN pooling dilution remains a liability even without global centrality inputs (Plain DQN: 96.0% vs. GNN-DQN: 84.0%), demonstrating that simple greedy heuristics outperform structural learning in this architectural configuration. Future work will address these directly: (1) evaluating hop counts and generalisation across multiple independently generated topologies of varying sizes and densities, (2) increasing the number of training/evaluation seeds to obtain a statistically conclusive comparison against the Random Forest baseline, (3) transitioning to per-node Q-value architectures and attention-based pooling (GAT) to preserve node-specific signals and enable native scaling to unseen network sizes, (4) time-varying contact graphs to better model genuine opportunistic connectivity, and (5) evaluation on real-world contact traces (e.g. CRAWDAD).

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